

Mathematics MCQ Test (30 Questions)

Topics: Algebra, Trigonometry, Functions, Limits, Continuity, Differentiation, Integration, Applications, Differential Equations, Linear Algebra

Q1. If $\log_2(x - 1) + \log_2(x + 1) = 3$, then the value of x is

- A) -3
- B) 3
- C) ± 3
- D) 5

Q2. The infinite geometric series $\sum_{n=0}^{\infty} 15 \left(\frac{1}{3}\right)^n$ converges to

- A) 15
- B) 22.5
- C) 30
- D) Diverges

Q3. If $\sin \theta + \cos \theta = \frac{1}{3}$, then $\sin 2\theta =$

- A) $-\frac{7}{9}$
- B) $\frac{7}{9}$
- C) $-\frac{8}{9}$
- D) $-\frac{8}{9}$

Q4. Evaluate $\tan\left(2 \tan^{-1} \frac{1}{3}\right)$.

- A) $\frac{3}{4}$
- B) $\frac{3}{5}$
- C) $\frac{5}{4}$
- D) $\frac{5}{8}$

Q5. Let $f(x) = \frac{x-2}{x+1}$. **Then** $f^{-1}(x) =$

- A) $\frac{x+2}{1-x}$
- B) $\frac{x+1}{x-2}$
- C) $\frac{x+2}{2-x}$
- D) $\frac{2-x}{x+1}$

Q6. Evaluate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1 - 3x}{x^2}$.

- A) 3
- B) $\frac{9}{2}$
- C) 9
- D) 6

Q7. $\lim_{x \rightarrow 0^-} \frac{|x|}{x}$ equals

- A) 1
- B) -1
- C) 0
- D) Does not exist

Q8. Evaluate $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$.

- A) $-\frac{1}{2}$
- B) $\frac{1}{2}$
- C) -1
- D) 1

Q9. $\lim_{x \rightarrow \infty} \frac{2x^3 + x}{x^3 - 1} =$

- A) 0
- B) 1
- C) 2
- D) ∞

Q10. For what value of k is

$$f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

continuous?

- A) 3
- B) 6
- C) 9
- D) 12

Q11. Which condition is sufficient to guarantee that $f(x) = 0$ has a solution in (a, b) ?

- A) $f'(x)$ exists
- B) $f(a) = f(b)$
- C) f continuous and $f(a)f(b) < 0$
- D) $f''(x) > 0$

Q12. If $y = \sin(x^2 e^x)$, then $\frac{dy}{dx}$ equals

- A) $\cos(x^2 e^x) (2x + x^2) e^x$
- B) $\sin(x^2 e^x) (2x + x^2) e^x$
- C) $\cos(x^2 e^x) (2 + x) e^x$
- D) $\cos(x^2) e^x$

Q13. If $x^2 + y^2 = 25$, then $\frac{dy}{dx} =$

- A) $\frac{y}{x}$
- B) $-\frac{y}{x}$
- C) $-\frac{x}{y}$
- D) $\frac{x}{y}$

Q14. The coefficient of x^3 in the Maclaurin expansion of e^x is

- A) $\frac{1}{6}$
- B) $\frac{1}{3}$
- C) $\frac{1}{2}$
- D) 1

Q15. If $f(x) = x^3 - 3x$, then f has a local maximum at

- A) $x = 1$
- B) $x = -1$
- C) $x = 0$
- D) $x = 3$

Q16. Evaluate $\int x e^x dx$.

- A) $e^x(x + 1) + C$
- B) $e^x(x - 1) + C$

C) $xe^x + C$

D) $e^x + C$

Q17. Evaluate $\int \sin^2 x \, dx$.

A) $\frac{x}{2} - \frac{\sin 2x}{4} + C$

B) $\frac{x}{2} + \frac{\sin 2x}{2} + C$

C) $-\cos x + C$

D) $\frac{1}{2} \sin 2x + C$

Q18. Evaluate $\int \frac{1}{x^2 - 1} \, dx$.

A) $\frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$

B) $\ln |x^2 - 1| + C$

C) $\frac{1}{x-1} + C$

D) $\tan^{-1} x + C$

Q19. If $F(x) = \int_1^{x^2} \cos t \, dt$, **then** $F'(x) =$

A) $2x \cos(x^2)$

B) $\cos(x^2)$

C) $2x \sin(x^2)$

D) $\cos x$

Q20. Determine $\int_1^{\infty} \frac{1}{x^2} \, dx$.

A) 0

B) 1

C) Diverges

D) ∞

Q21. $\int_0^{\pi/2} \cos^2 x \, dx =$

- A) 0
- B) $\frac{\pi}{4}$
- C) $\frac{\pi}{2}$
- D) 1

Q22. Area enclosed by $y = x$ and $y = x^2$ is

- A) $\frac{1}{6}$
- B) $\frac{1}{3}$
- C) $\frac{1}{2}$
- D) 1

Q23. Solve $\frac{dy}{dx} = 3y, y(0) = 4.$

- A) $y = 4e^{3x}$
- B) $y = 3e^{4x}$
- C) $y = e^{3x}$
- D) $y = 12e^x$

Q24. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$. Then $\det(A) =$

- A) 0
- B) 10
- C) 24
- D) 120

Q25. The eigenvalues of $A = \begin{pmatrix} 4 & 1 \\ 0 & 2 \end{pmatrix}$ are

- A) 4 and 1
- B) 4 and 2
- C) 2 and 1
- D) 5 and 1

Q26. If A is an $n \times n$ matrix with $\det(A) = 0$, then

- A) A is invertible
- B) $Ax = 0$ has only the trivial solution
- C) $Ax = 0$ has a nontrivial solution
- D) $\text{rank}(A) = n$

Q27. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. Then $A^{-1} =$

- A) $\begin{pmatrix} -2 & 1 \\ 3 & -1 \\ 2 & -2 \end{pmatrix}$
- B) $\begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$
- C) $\begin{pmatrix} -2 & 1 \\ 3 & 1 \\ 2 & 2 \end{pmatrix}$
- D) $\begin{pmatrix} 2 & -1 \\ 3 & 1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$

Q28. The rank of $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$ is

- A) 1
- B) 2
- C) 3
- D) 0

Q29. The null space of $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ is spanned by

- A) $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$
- B) $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$
- C) $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
- D) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q30. If A is a 3×3 matrix with $\det(A) = 5$, then $\det(2A) =$

- A) 10
- B) 20
- C) 40
- D) 5